

Investigation into the Biodiversity Management and Conservation Practices in Southeast of Nigeria

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Abstract

The richness or sparseness of biodiversity within any ecosystem tells a story of the socio-economic, wealth and value systems of the inhabitants. The aim of this study is to investigate the conservation and management of biodiversity within the five Southeastern States in Nigeria through mapping of the forest covers changes and the areas where biodiversity has not been or have been interfered by human activities. The Biodiversity Intactness Index (BII) which has been adopted worldwide as a standard metrics for the monitoring of biodiversity states within any ecosystem across the tropical forest belts was employed in this study alongside percent forest cover, forest cover loss and population density data all gotten from UN Biodiversity Lab. The result from the BII recorded a 93% mean which is above the proposed safe limit of 90%, this indicates that the Southeastern States of Nigeria has done fairly well in managing their biodiversity by maintaining a relative richness in forest cover diversity within the ecosystem. It was also noted that areas within the >50% forest cover are along River channels notably protected by the locals as sacred forests. The result of the correlation analysis **-0.455**, shows a negative relationship between BII and Population density. Thus, in order to maintain the state of the biodiversity and also improve on it, local conservation policies need to be enacted in line with the peoples' culture and belief in sacred forests to effectively conserve the ecosystem which is vulnerable to the teeming human population.

Keywords: *Biodiversity, Conservation, Population, UN Biodiversity Lab and Remote Sensing*

Introduction

Biodiversity is the spatial distribution, varieties and abundance of life within a given ecosystem. The biodiversity of an area performs an ecosystem function as a life supporting system (Rawat & Agarwal, 2015). This is because any area within the earth planet that is void of any form of biodiversity denotes a location where life cannot thrive. Hence it is paramount to manage and conserve any form of biodiversity wherever they are found.

Most of earth's floras which form part of the biodiversity have undergone various unsustainable changes overtime mostly induced by either natural or anthropogenic factors. Though most of these floras have the ability to repopulate overtime, some have gone into extinction (Singh, 2010) thus bringing about some element of imbalance in the ecosystem. In recent times, climate change, forest fires and unsustainable land uses have been identified as the major factors that are affecting vegetation covers around the world (Yunfeng, Rina, & Yang, 2019; Kleeman, Baysal, Bulley, & Fürst, 2017). It has been argued that climate change which is as a result of excess CO₂ in the atmosphere has led to fast regeneration due to 'fertilization effect' on the various vegetation covers around the world (Franks, *et al.*, 2013) but most scholars have argued against this notion (Rawat & Agarwal, 2015). Undeniably, human interferences have been pinpointed to be causing much of damages to the floras and the entire biodiversity as being recorded in the world today (Kleeman, Baysal, Bulley, & Fürst, 2017). Invariably the increasing human population creates more pressure on the already declining land resources and this leads to more specie extinctions and loss of the valuable ecosystem services. The richness in species diversity within an ecosystem tells a story of how well the ecosystem is being managed and conserved. It further tells a story of the people and the values they place on the ecosystem in terms of economic, cultural and social importance.

Many policies have been outlined by different global agencies to manage and conserve the biodiversity on earth such as; International legal framework for biodiversity protection, Convention on Biodiversity (CBD), The United Nations Convention on the Laws of the Seas, World Heritage Convention, the RAMSAR Convention on the Conservation of Wetlands of International Importance especially Waterfowl Habitat, the United Nations Conference on Climate Change and Convention on International Trade In Endangered Species of Flora and Fauna but their combined effects are yet to be felt in Africa. In order to effectively and efficiently monitor these changes within the flora community in line with these policies, various algorithms have been developed as to that effect such as traditional in-situ field observations and through the use of remotely sensed imageries from the various satellite that have been designated to carry out such activities. For this study, we utilized the UN Biodiversity Lab spatial data that is freely available to look into the conservation and management of biodiversity within the five (5) Southeastern States in Nigeria in order to understand the changes that have occurred on the vegetation covers since the 21st century and the areas where biodiversity is stressed due to human impacts on them.

The UN Biodiversity Lab is an interactive online mapping program that supports informed decision-making on biodiversity conservation and development challenges. Its source of spatial data ranges from UN agencies, the World Conservation Monitoring Centre (WCMC), the Global Resource Information Database (GRID-Geneva), to the US National Aeronautics and Space Administration (NASA) and others. The UN Biodiversity Lab which is powered by MapX for an interactive visualization and map making is solely funded by Global Environment Facility (GEF).

Methodology

Study area

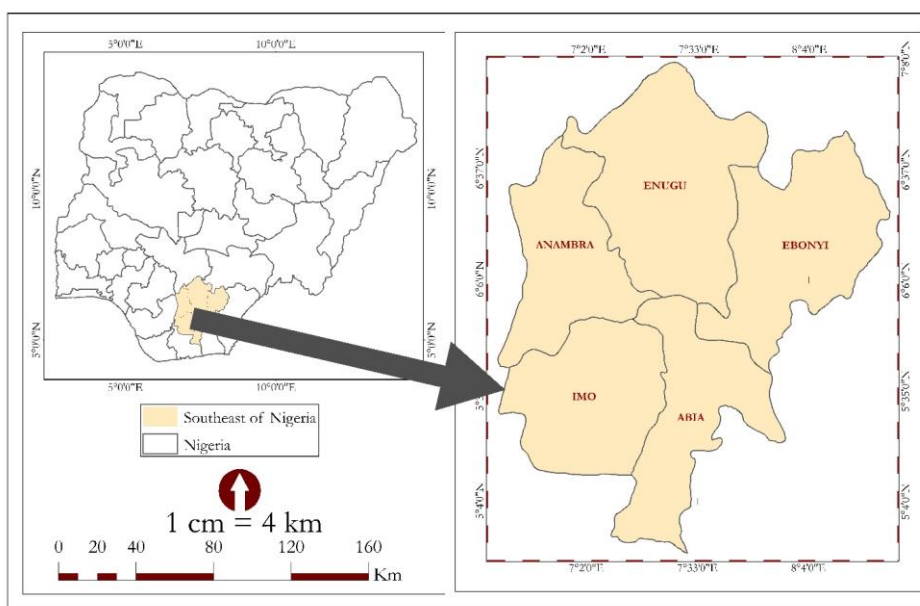
The study location which comprises the five Southeastern States in Nigeria; Imo, Abia, Anambra, Enugu and Ebonyi, lies within Longitude 7°2'0"E to 8°4'0"E and Latitude 5°4'0"N to 7°8'0"N. The total area coverage is about 28,713.987 square kilometer (sqkm) which accounts for 3.15% (approx.) of the about 912,038.625sqkm total land area of Nigeria. The people of Southeastern Nigeria are predominantly Igbo, with the majority speaking Igbo and a minority speaking Igala. They are renowned for their commercial business acumen.

The study area falls within the Tropical Rainforest Ecological Zone which is classified as α sub-humid according to Koppen's Aw climate classification (Ayode, 2004). The Eco-Zone is characterized by high rainfall amount of about 1500-2500mm and moderate temperature ranges of about 26-28°C, and good sunshine hours of about 9 hours daily. The eco-zone experiences two major seasons; rainy which occurs from March to October every year with its peak in July-September with a little a break in August and dry seasons occurs from November to February annually with its peak from December to January. These seasons are controlled by the movement of the Inter-Tropical Front (ITF) or Inter Tropical Convergence Zone (ITCZ) or the Inter Tropical Discontinuity (ITD) that was formed by the two airmass acting on the zone; the Tropical Continental (cT) airmass from Sahara Desert and the Tropical Maritime (mT) airmass from the Atlantic Ocean.

The study area is made up of mosaic of different vegetation cover and most have been altered overtime. The vegetation cover ranges from light rainforest, mangrove and swampy forest within the southern part of the study area, dense vegetation cover within the highlands and traces of savannah vegetation towards the northern part of the study area. Riparian vegetation covers are also found along the stream channels and some of them are being preserved. These preserved vegetation around the stream channels are known as 'OfiaMmili' (forest of the river/stream) by the natives and are not to be tampered with until recent times. Some of the dense vegetal covers are found within shrine grooves are dedicated to the Alusi (demi-god), worshipped and conserved. They are known as 'Ofia Alusi' (forest of the god) or in some cases they go by the name of the Alusi e.g., 'Ofia Udo' (forest of Udo god) and out of bounds from human intrusions.

Within these confined grooves, the typical resemblance of the equatorial rainforest vegetation can be seen with its characteristics of tall trees with wide girths and broad leaves e.g., Iroko tree, Baobab and other tree cover species.

Inherently, human activities such as bush burning, agriculture and urban development poses a great threat to the natural vegetation in the area and in most cases have led to the altered state of this vegetation.



Data description

The raster datasets were all gotten from the UN Biodiversity Lab and further post-processed in a GIS software environment basically for clipping to study area shape and map embellishment. Below are the algorithms guiding the raw data processing as carried out by different authors.

Biodiversity Intactness Index (2016)(BII) is an indicator best situated for the tropical forest to give the average richness- and area-weighted impact of a set of activities on the populations of a given group of organisms within a specific ecosystem. The population impact (I_{ijk}) is defined as the populations of species group i under land use activity k in ecosystem j , relative to a reference population in the same ecosystem type. The BII is calculated as:

$$BII = \frac{\sum_i \sum_j \sum_k R_{ij} A_{jk} L_{ijk}}{\sum_i \sum_j \sum_k R_{ij} A_{jk}}$$

Where R_{ij} = richness (number of species) of taxon i in ecosystem j , and A_{jk} = area of land use k in ecosystem j (Scholes & Biggs, 2005; Newbold *et al.*, 2016).

Population Density (2020) was extracted from the UN Biodiversity Lab having been sourced from Gridded Population of the World, Version 4 (GPWv4). Population Density, Revision 10 consists of estimates of human population density (number of persons per square kilometre) based on counts consistent with national censuses and population registers, for the years 2000, 2005, 2010, 2015, and 2020. A proportional allocation gridding algorithm, utilizing approximately 13.5 million national and sub-national administrative units, was used to assign population counts to 30 arc-second grid cells. The population density raster was created by dividing the population count raster for a given target year by the land area raster. The data files were produced as global raster at 30 arc-second (~1 km at the equator) resolution.

Forest Cover (2000) data set is a collaboration between the GLAD (Global Land Analysis & Discovery) lab at the University of Maryland, Google, USGS, and NASA, and shows the percent tree cover (0-100) in year 2000 for all global land areas (outside of the polar regions) at 30 × 30-meter resolution. Tree cover is defined as the tree canopy cover of all vegetation over 5 metres in height. The data is derived from analysing over 600,000 multispectral Landsat 7 EMT+ satellite images in Google Earth Engine (Hansen, *et al.*, 2013).

Forest Cover Loss (2000-2018) this data set is a disaggregation of total 'Forest Loss' to annual time scales and is a collaboration between the GLAD (Global Land Analysis & Discovery) lab at the University of Maryland, Google, USGS, and NASA. The data indicates the year where forest loss primarily occurred (based on year 2000 forest cover) for all global land areas (outside of the Polar Regions) at 30×30-meter resolution. Forest loss was disaggregated to annual time scales using a set of heuristics derived from the maximum annual decline in percent tree cover and the maximum annual decline in minimum growing season Normalized Vegetation Difference Index (NDVI). Trends in annual forest loss were derived using an ordinary least squares slope of the regression of y =annual loss versus x =year. The data is derived from analysing over 600,000 multispectral Landsat 7 EMT+ satellite images in Google Earth Engine (Hansen, *et al.*, 2013).

Result and Discussion

Figure 1 depicts the percent tree canopy cover density of vegetation above 5 metres in height at 30*30m resolution within the study area in year 2000. The 25-50% tree cover category is found almost everywhere in the study area but it is more prominent within the Northern part of the area at Enugu, Ebonyi and most part of Anambra State. Within this category we have the banana, oil palm trees, rubber, or *Eucalyptus* plantations and cultivated cash crops farmlands, these 'artificial forests' poses a great threat to the tropical biodiversity. The encroachment of these tree species and their rate of colonization within the tropical forest don't show good forest conservation and management but rather a lack of it. This is because of their 'Monoculture'

which does not reflect the richness in biodiversity, climate regulation, carbon storage, water supplies amongst other ecosystem services a typical tropical forest represents (Tropek, *et al.*, 2014). Their presence also points towards a forest disturbance.

The >50% tree canopy cover are more dominant along surface water channels most of these categories are still in their natural state where they are being intentionally preserved by the locals due to their cultural beliefs and most have been altered albeit a little. This forest category is found along the river/stream channels are generally known as Riparian forests.

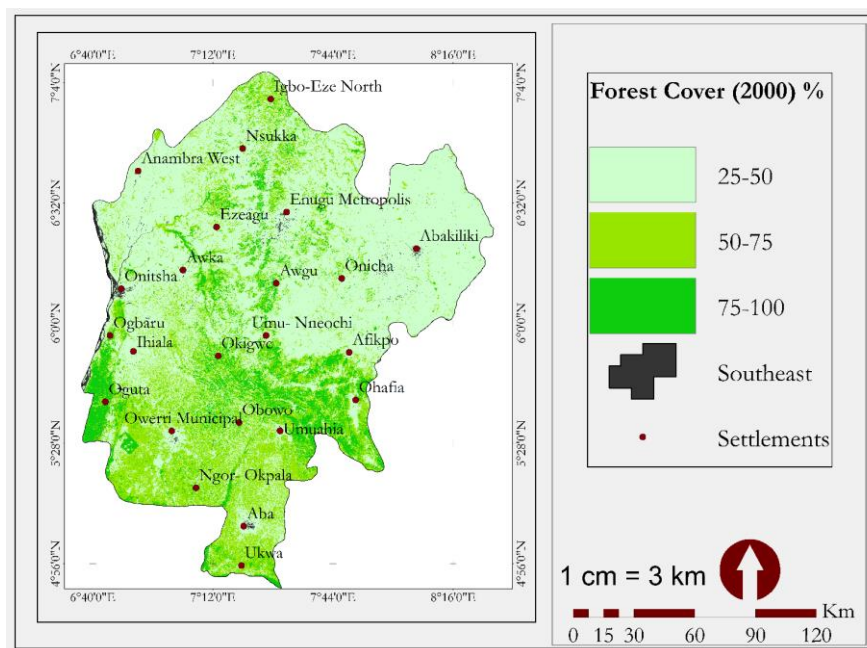


Figure 1: Percent Forest Cover in year 2000

The forest cover loss based on the percent tree canopy cover of the year 2000 shown in figure 2 and further overlaid on the 2000-year forest map as shown in figure 3 distinctly reveals to us where deforestation, human activities, bush burning and land clearing have greatly caused forest stand loss from 2000 (yellow) to 2018 (red). The in-between years are represented with orange colour to show which areas that were affected within those years.

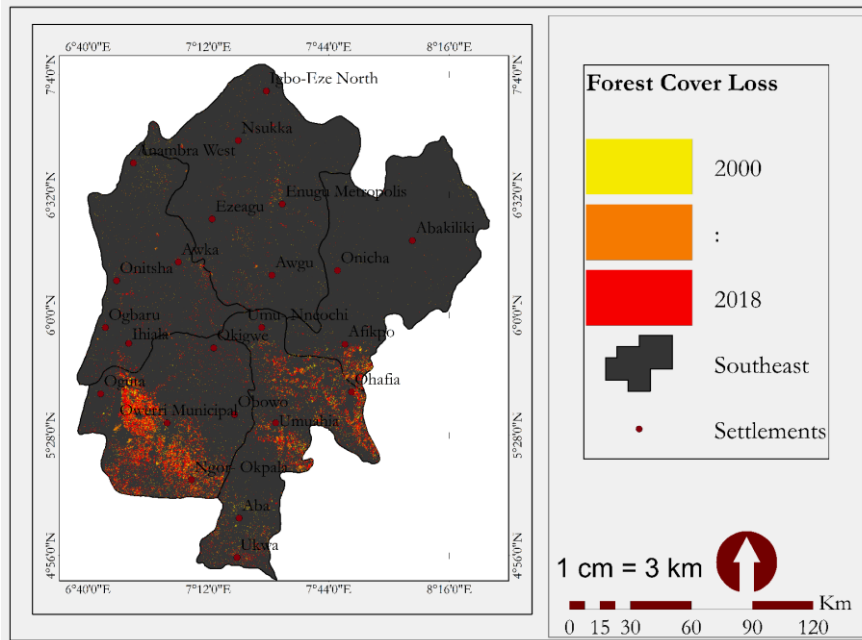


Figure 2: Forest Cover Loss 2000-2018

Over the past 18 years, Abia and Imo States most notably around their urban centers of Aba, Umuahia and Owerri had suffered a huge loss in its forest cover. These losses occurred across the entire percent tree canopy cover category but mostly within the >50% tree canopy cover. The <50% tree canopy cover fared well all through the 18 years with little or no changes occurring within them. Most of the disturbances that occurred within the >50% tree cover were orchestrated by influx of population within the urban areas and this can be related to population pressures on the few land resources.

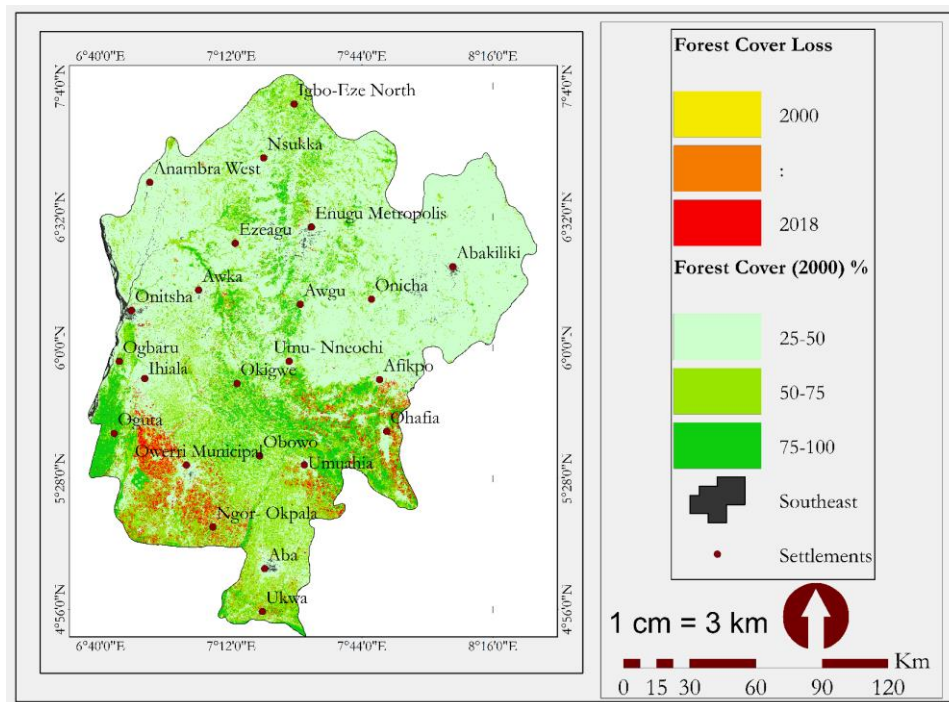


Figure 3: Overlay of Forest cover loss (2000-2018) on Forest cover (2000)

The 2020 projected population density in figure 4 further reveals the human pressures on the few land resources per SqKm. From the map the highest population density is concentrated at the two major commercial cities within the Southeastern part of Nigeria which is Onitsha and Aba. This is closely followed by other major cities, Awka, Enugu Metropolis, Umuahia, Owerri then we have the satellite towns like Nsukka, Obollo-afor all in the Northern tip of Enugu, following in the moderate persons per SqKm. The lowest persons per SqKm of less than 1000 are concentrated within the Ebonyi, Enugu and fewer locations within Anambra, Imo and Abia States.

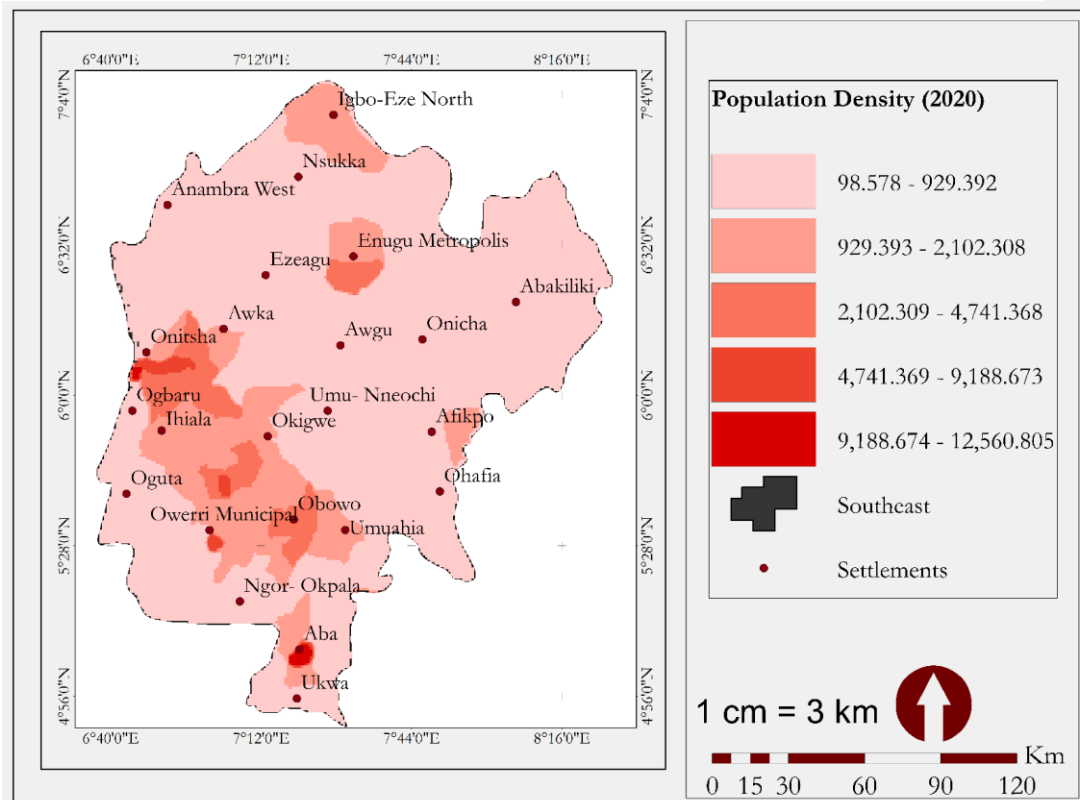


Figure 4: Population Density 2020

The BII has been proposed as a metrics that should be used in assessing the states of these biodiversity both at local and global scale. The BII result presented in figure 5 reveals the areas where our biodiversity is under stress due to the rising human population. The colour gradation is from a shade of red to yellow which indicates ‘highly stressed’ or reduced forest cover density (67 to 87%) while from green to deep blue indicates high forest cover density under a stable state (90 to 105%). Following the 2016 estimated global mean of BII as 84.6% (Newbold *et al.*, 2016) which also falls below the proposed 90% safe limit by Steffen *et al.*, (2015) our study area passed the safe limit with an estimated mean of 93%. This indicates that our biodiversity is fairly managed and stable with a fair abundance offorest cover. Furthermore, on a mappublished by (Battistella, *et al.*, 2018) the region is noted to fall within the 5-10% coverage of the terrestrial ecoregions of the world, excluding waterbodies and other inert objects being protected areas - calculated at the national level. Most of the few areas where the BII indicated stressed

biodiversity (red to yellow colour) falls within the areas of high population density such as the major cities of Onitsha, Aba, Awka, Enugu Metropolis, Abakiliki, Umuahia and Owerri.

Furthermore, a correlation analysis was carried out to determine the type of relationship that exists between the Biodiversity Intactness and Population density. The analysis gave us a **-0.455** which indicates a negative relationship between the two variables. This further buttress the fact that the increasing human population poses great threat to the biodiversity at large.

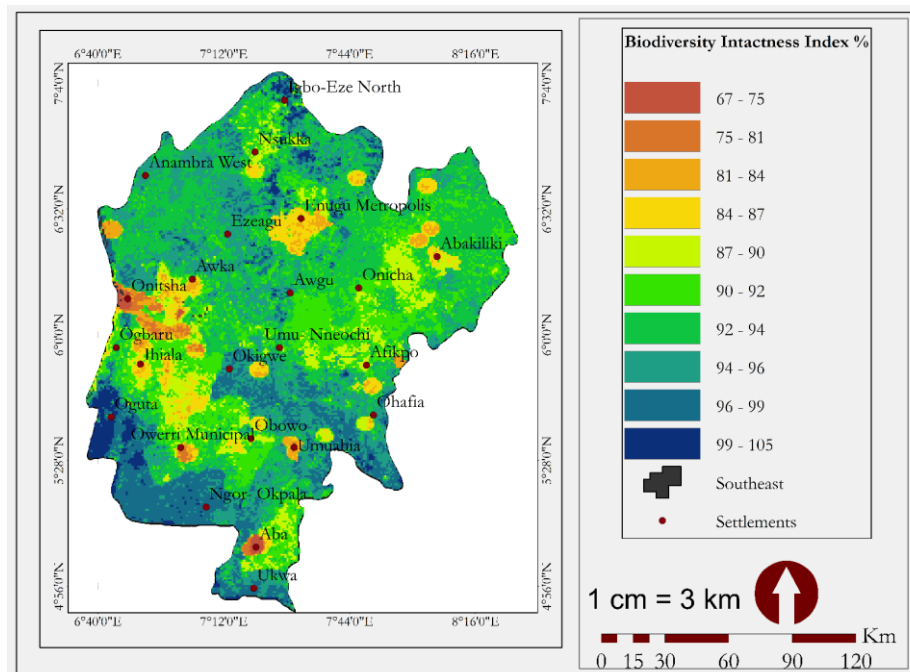


Figure 5: Biodiversity Intactness Index

Conclusion

With a BII mean of 93%, the Southeastern States of Nigeria seems to have done fairly well in the conservation and management of their biodiversity but more proactive measures need to be taken to address the conservation especially in urban areas that are threatened by population growth. The teeming human population within the urban centers and the quest for more lands to satiate the ever-brimming needs of this urban population creates a huge threat to the biodiversity. The result of the correlation analysis which yielded a **-0.455** further reveals the negative relationship between the two variables under study and this call for concern from the various

State stakeholders. The State stakeholder's needs to enact local biodiversity conservation policies that would aim at further improving the state of biodiversity within the States and the protection of the same from the ever-rising human population by leveraging on the peoples' cultural belief in sacred forests and respect of such. Thus, it is paramount to align forest conservation policies in line with the people's traditional and cultural beliefs as they have proven over time to be more effective in biodiversity conservation (examples, conservation of Awka monkeys, Agulu lake crocodile family and African pythons in some part of Anambra State due to their inferred sacredness by the locals) than the existing global policies.

The use of BII has proven to be a good indicator in monitoring of the biodiversity and the conservation processes. The UN Biodiversity Lab web tool has become a necessary tool when analyses pertaining to the biodiversity needs to be done due to its huge archive and access to world standard analyses in biodiversity management. The web tool is powerful for an effective and efficient biodiversity conservation and management at different scales.

Acknowledgement

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Disclaimer: All the data used in this work was sourced from UN Biodiversity Lab at www.unbiodiversity.org

References

- Ayode, J. (2004). *Introduction to Climatology for the Tropics*, second edition. Ibadan: Spectrum Books Limited.
- Battistella, L., Mandrici, A., G., D., Bertzky, B., Bastin, & Dubois, G. (2018). *Terrestrial Ecoregion Protection 2018 (National Level)*. European Union.
- Franks, P. J., Adams, M. A., Amthor, J. S., Barbour, M. M., Berry, J. A., Ellsworth, D. S., . . . Caemmerer, S. v. (2013). Sensitivity of plants to changing atmospheric CO₂ concentration: from the geological past to the next century. *New Phytologist*, 197, 1077-1094.
- Hansen, M. C., Potapov, P. V., Moore, R., Hancher, M., Turubanova, S. A., Tyukavina, A., . . . Townshend, J. R. (2013). High Resolution of Global Maps of 21st Century Forest Cover Change. *Science* 342, 850-853.
- Kleeman, J., Baysal, G., Bulley, H. N., & Fürst, C. (2017). Assessing driving forces of land use and land cover change by a mixed-method approach in north-eastern Ghana, West Africa. *Journal of Environmental Management*, 196, 411-442.
- Newbold, T., Hudson, L. N., Arnell, A. P., & Contu, S. (2016). Global Map of Biodiversity Intactness Index. *Science* 353, 288–291.
- Obiadi, I., Nwosu, C., Ajaegwu, N. E., Anakwuba, E. K., Onuigbo, N. E., Akpunonu, E. O., & Ezim, O. E. (2011). Gully Erosion in Anambra State, South East Nigeria: Issues and Solution. *International Journal of Environmental Sciences, Volume 2, No 2*, 795-805.
- Rawat, U., & Agarwal, N. (2015). Biodiversity: Concept, threats and conservation. *Environment Conservation Journal* 16(3), 19-28.
- Scholes, R. J., & Biggs, R. (2005). A biodiversity intactness index. *Nature* Vol 434; DOI: 10.1038/nature03289 · Source: PubMed, 45-49.
- Singh, A. (2010). Biodiversity Conservation: Protecting Species before they are gone. *Science Reporter* retrieved from <https://www.researchgate.net/publication/279182331>, 1-43.
- Steffen, W., Richardson, K., Rockström, J., Cornell, S. E., Fetzer, I., Bennett, E. M., . . . Sörlin, S. (2015). 'Planetary boundaries: Guiding human development on a changing planet'. *Science*, 347(6223).
- Tropek, R., Sedláček, O., Beck, J., Keil, P., Musilová, Z., Šimová, I., & Storch, D. (2014). Comment on “High-resolution global maps of 21st-century forest cover change”. *Forest Surveys, Technical Comment Vol 344 Issue 6187* from sciencemag.org, 981-d.
- Yunfeng, H., Rina, D., & Yang, H. (2019). Vegetation Change and Driving Factors: Contribution Analysis in the Loess Plateau of China during 2000–2015. *Sustainability*, (11) 1320, 1-16.